Poiseuille flow

Wiki *Enhanced dissipation for passive scalars*

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Page team: to be named · Status: skeleton
Discuss on Zulip: #examples, topic “Poiseuille flow”.

[Lead: what changes compared with Couette flow, and why.]

1 Setting

[The flow, the domain, the critical point, what decays.]

2 The result

Theorem 1 (enhanced dissipation for Poiseuille flow)**.** *[The statement.]*

Lean: not yet formalized.

Idea. *[The key mechanism in a few lines.]*

Proof architecture. *[The proof as a short chain of steps, and its diagram.]*

3 What changes compared with Couette flow

For each technique: what carries over from Couette flow, what is lost, and the proof for this example.

3.1 Explicit solution

[What is lost.]

3.2 Spectral methods

Proof by spectral methods. *[The proof.]* $\square$

3.3 Hypocoercivity

Proof by hypocoercivity. *[The proof.]* $\square$

3.4 Stochastic methods

Proof by stochastic methods. *[The proof.]* $\square$

3.5 Hörmander theory

Proof by Hörmander theory. *[The proof.]* $\square$

References